

5.H1 Definition of Congruence in Terms of Rigid Transformations

Lesson Introduction

 Benchmark	MA.912.GR.2.7 Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations.		 Learning Target	I can use the definition of congruence to justify that SSS congruence, SAS congruence, ASA congruence, and AAS congruence are sufficient to show that two triangles are congruent.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	How can you use the definition of congruence to justify that SSS, SAS, ASA, and AAS congruence are sufficient to show that two triangles are congruent?			
 Lesson Narrative	In this lesson, students will extend their learning of congruence in terms of rigid transformations. Students will learn how to justify congruence through a structured exploration using various rigid transformations. Through the process of the different transformations, students will explore the connection between their patty paper verification and a series of transformations to understand that the definition of congruence in terms of rigid transformations shows that SSS, SAS, ASA, and AAS congruence are sufficient.			
 Component Summary	5.H1.1 Warm-Up (~5 min.)	Geometry as an animal activity (<i>SE p. 257</i>)	5.H1.2 Exploration (30-35 min.)	Justifying congruence activity (<i>SE p. 258</i>)
	5.H1.3 Wrap-Up (~5-10 min.)	Square, triangle, circle reflection activity (<i>SE p. 261</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Patty Paper	
			Additional Preparation	
			This lesson is designed for students enrolled in honors Geometry as the benchmark addressed is only included in the Geometry Honors course.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly state the transformations that map a triangle onto another. - Students may not connect that the given information is shown to be true by showing that the definition of congruence was applied by mapping the triangles. - Students may conclude that two triangles are congruent without proper evidence.	The primary function of this lesson is intended to be an exploration through Activity 5.H1.2. It is not a lesson to give students information, but rather to have them explore the concept to draw the conclusion on their own through the activity. Consider going through this activity on your own beforehand to look for potential points of misconception that may occur.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Throughout Activity 5.H1.2, students will get the opportunity to engage in individual processing time and work prior to structured discussion with a partner about their findings and conjectures.	MA.K12.MTR.5.1: Patterns and Structure This lesson is an extension of what students have learned about congruence in this unit. Students will relate those previously learned concepts to new concepts, while looking for similarities among problems.
 Differentiate Instruction	While this is an honors lesson, student grouping/partnering should be carefully considered based on performance throughout the unit up to this lesson. Some students may be in need of additional scaffolding on support. This lesson provides a great opportunity to clear up misconceptions about congruence and/or specific postulates and theorems.	
Lesson Notes:		

5.H1.1 Warm-Up: Geometry as an Animal

Component Overview: Students will complete two open-ended questions about geometry as an animal.



5.H1.1 Warm-Up: Geometry as an Animal

1. If geometry were an animal, it would be a ...

Answers will vary. If geometry were an animal, I think it would be a turtle.

because ...

Both are majestic and intentional. Turtles do have geometric patterns on their shell. Turtles like Geometry must make every step carefully and intentionally.

2. It would live in ...

Answers will vary. I would live in the ocean.

because ...

We all know it is out there, however when we do see it, we are surprised and inspired.



This activity is designed to get students thinking and talking. Consider having students engage in a Think-Pair-Share routine with this activity. Allow students some individual think time and then ask them to work with a partner to share their thoughts and answers with one another.

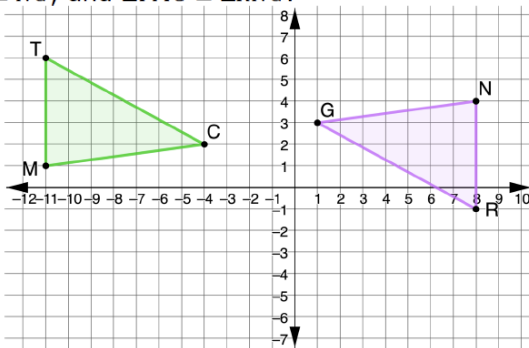
5.H1.2 Exploration: Justifying Congruence

Component Overview: Students will explore the concept of congruence further to learn how to justify congruence through three, multi-part, scaffolded questions. For this activity, students will need a partner.



5.H1.2 Exploration: Justifying Congruence

1. Triangles TCM and RGN are shown, where $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$.



- Discuss with your partner what can be concluded from the given information.
Answers will vary. I can conclude that the triangles are congruent by SAS.
- Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Conclusion	My Summary of Their Reason	Initials



When circulating during this activity, listen for academic language. If students are making general observations (such as “one is green and one is blue”), allow those observations, but offer a question that gets them to think about their location or how they may be related to one another.



Add a layer of physical movement to this activity. Consider having students find at least two different students by getting up and moving around the room and finding two unique answers. Consider the use of a timer to maintain instructional momentum and to ensure students don't get lost in the logistics of this activity.

5.H1.2 Exploration: Justifying Congruence (continued)

- c. What rigid motions can be applied so that $\triangle TCM$ is mapped to $\triangle RGN$?
Answers will vary. Rotation 180° about the origin and then translated left 3 and up 5.
- d. Discuss with your partner what can be concluded from the rigid motions.
Answers will vary. The shapes are congruent.
- e. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Conclusion	My Summary of Their Reason	Initials

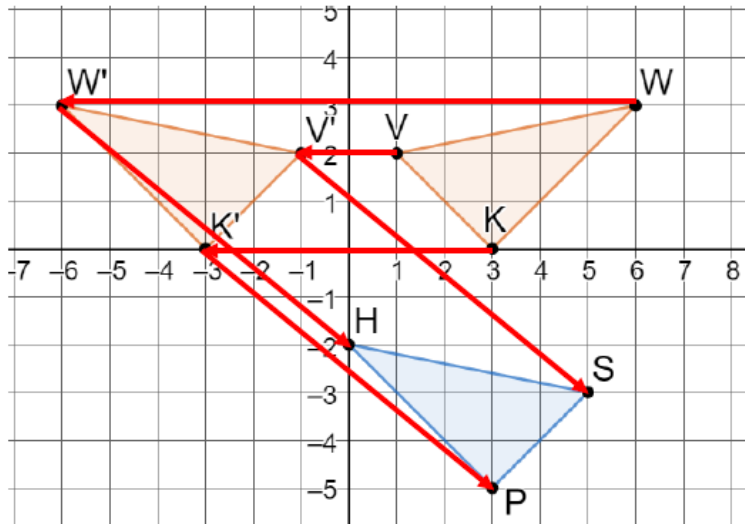
- f. Discuss with your partner why showing that the triangles are congruent using the definition of congruence in terms of rigid transformations shows that SAS congruence is sufficient to prove the triangles are congruent. Summarize your discussion.
Answers will vary. Because one triangle was mapped to another using rigid transformations the definition of congruence states that two triangles are congruent. Since $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$ were given as true then SAS congruence is sufficient to prove the triangles are congruent.



For question 1e, consider adding an additional layer that the two students they use for this table can't be the same two from the earlier part. This will help students engage in diversity of thought and perspective.

5.H1.2 Exploration: Justifying Congruence (continued)

2. Triangles VWK and SHP are shown. Assume $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$.



- a. Explain how you could use patty paper to verify $\triangle VWK \cong \triangle SHP$.

I could use patty paper to trace the triangle. Next, I can line the tracing up with the other image to prove that they are congruent.

- b. Write a series of transformations that will map $\triangle VWK$ onto $\triangle SHP$.

Answers will vary. Reflect $\triangle VWK$ across the y-axis, then translate it down 5 right 6.

- c. Show each on the coordinate grid.
- Draw $\triangle V'W'K'$, the result of the first transformation
 - Map the vertices of $\triangle VWK$ to $\triangle V'W'K'$
 - Map the vertices of $\triangle V'W'K'$ to $\triangle SHP$
- Shown on the graph*



For question 2, consider providing students with patty paper so they can use it with question 2a. Instead of just explaining how they could use it to verify, actually ask them to verify it using patty paper.



There are multiple answers that could be correct for question 2b. For students who are moving through the activity quickly, challenge them to come up with an additional series of transformations that could produce the same results.



Consider providing students with a coordinate grid or graph paper to complete question 2c. Otherwise, students may attempt to “sketch” this and lose sight of accuracy.

5.H1.2 Exploration: Justifying Congruence (continued)

d. Complete the statements.

The outcome of mapping the transformations of

$\triangle VWK$ to $\triangle V'W'K'$ then from $\triangle V'W'K'$ to $\triangle SHP$ shows

that V' coincides with $\begin{array}{l} \circ H, \\ \circ S, \\ \circ P, \end{array}$ W' coincides

with $\begin{array}{l} \circ H, \\ \circ S, \\ \circ P, \end{array}$ and K' coincides with $\begin{array}{l} \circ H, \\ \circ S, \\ \circ P. \end{array}$

Because the definition of $\begin{array}{l} \circ \text{congruence} \\ \circ \text{similarity} \end{array}$

in terms of rigid motions preserves distance and

angles, triangle VWK is $\begin{array}{l} \circ \text{congruent} \\ \circ \text{similar} \end{array}$ to triangle

$\triangle SHP$. Since $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$

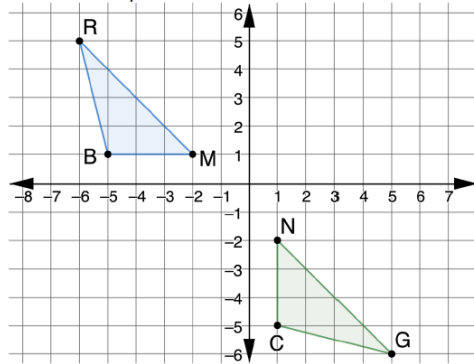
was given, $\begin{array}{l} \circ \text{AAS} \\ \circ \text{ASA} \\ \circ \text{SAS} \\ \circ \text{SSS} \end{array}$ is sufficient to show triangles
are congruent.



When circulating the room, consider asking students to rephrase what these statements are saying in their own words. Often, students will complete this type of question line by line and not go back to review the concept within the context provided.

5.H1.2 Exploration: Justifying Congruence (continued)

3. Triangles RMB and GNC are shown, where $RM \cong GN$, $m\angle BRM \cong m\angle CGN$, and $m\angle BMR \cong m\angle CNG$.



Adriana knows that there is a series of rigid motions that maps $\triangle RMB$ onto $\triangle GNC$.

- a. Select all true statements after the transformations.
- ☐ Line segment RB coincides with line segment GN .
 - ☐ Line segment BM coincides with line segment CN .
 - ☐ Line segment CG coincides with line segment BR .
 - ☐ Angle NCG coincides with angle RMB .
 - ☐ Angle RBM coincides with angle GCN .

- b. Explain how the given information and the definition of congruence in terms of rigid motions shows that $\triangle RMB \cong \triangle GNC$.

Answers will vary. It was given that two angles and an included side of one triangle are congruent to two angles and an included side of another triangle. Because the triangles were mapped to another using rigid transformations the definition of congruence states that two triangles are congruent. This shows that ASA is sufficient to show two triangles are congruent.



Consider using a digital tool, such as GeoGebra or Desmos, to allow students to manipulate these triangles in order to explore them and develop their answers to question 3.



When closing out this component, consider having a whole-group discussion on the answer to question 3b, as it anchors the activity conceptually.

5.H1.3 Wrap-Up: Reflection

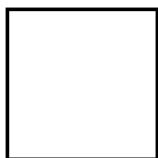
Component Overview: Students will complete three questions about what they've squared away, three important points, and something circling in their brain.



5.H1.3 Wrap-Up: Reflection

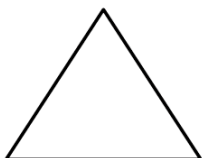
1. What is something that is now “squared away” in your mind about the definition of congruence in terms of rigid transformations?

Answers will vary.



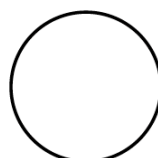
2. What are three “points” that are important for showing triangle congruence using the definition of congruence in terms of rigid transformations?

Answers will vary.



3. What is something that is still “circling” your brain about the definition of congruence in terms of rigid transformations?

Answers will vary.



The information provided from the student here, coupled with what was observed during the exploration, can provide information for differentiating instruction in later lessons.



Consider pairing students to discuss the things still circling their brains to see if peers can answer any of their questions or provide clarification.